

Q1. [10 pts] Let

$$f(x, y) = \begin{cases} \frac{x^2 + y^4}{1 - \sqrt{x^2 + y^4 + 1}} & \text{if } (x, y) \neq (0, 0), \\ -2 & \text{if } (x, y) = (0, 0). \end{cases}$$

Show that f is continuous at $(0, 0)$.

Q2. [10 pts] Let $f(x, y, z) = \frac{x \cos(x^2 + y)}{yz}$.

(a) Find the gradient of f .

(b) Find the unit vector $\mathbf{u}$ such that $D_{\mathbf{u}}f(1, -1, 2)$ is maximal.

Q3. [10 pts] Find the extreme values of the function $f(x, y, z) = x^2y^2z^2$ subject to the constraint $x^2 + y^2 + z^2 = 1$.

Q4. [10 pts] A wire takes the shape of the arc of the parabola $y^2 = 4x$ between the points $(0, 0)$ and $(1, 2)$. Find the mass of the wire if its linear density is $\rho(x, y) = xy$.

Q5. [10 pts] Use Green's Theorem to find the area of the region enclosed by the cardioid $r = 1 - \cos \theta$, $0 \leq \theta \leq 2\pi$.

Q6. [10 pts] Let $\mathbf{F}(x, y, z) = -\frac{2xz}{(x^2 + 2y^2 + 1)^2} \mathbf{i} - \frac{4yz}{(x^2 + 2y^2 + 1)^2} \mathbf{j} + \frac{1}{x^2 + 2y^2 + 1} \mathbf{k}$.

Use the Fundamental Theorem for Line Integrals to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is any smooth curve from $A(0, 0, 1)$ to $B(2, 1, 4)$.

Q7. [10 pts] Let S be the surface $z = x^2 + y^2$, $z \leq 4$ with upward orientation, and let $\mathbf{F}(x, y, z) = \langle yz, -xz, e^{xyz} \rangle$. Use Stokes' Theorem to find $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$.

Q8. [15 pts] Let E be the solid in the first octant that is bounded by the coordinate planes and the plane $x + y + z = 1$, and let S denote the boundary surface of E with outward orientation. Use the Divergence Theorem to evaluate the flux of the vector field $\mathbf{F}(x, y, z) = \langle xy, y^2, yz \rangle$ across S .

Q9. [15 pts] Let S be the surface with vector equation

$$\mathbf{r}(u, v) = \langle (2 + \cos v) \cos u, (2 + \cos v) \sin u, \sin v \rangle, \quad -\pi \leq u, v \leq \pi.$$

(a) Find an equation for the tangent plane to S at the point $(3, 0, 0)$.

(b) Find the surface area of S .

Q1. The function f is continuous at $(0, 0)$ since

$$\begin{aligned}\lim_{(x,y) \rightarrow (0,0)} f(x,y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^4}{1 - \sqrt{x^2 + y^4 + 1}} \cdot \frac{1 + \sqrt{x^2 + y^4 + 1}}{1 + \sqrt{x^2 + y^4 + 1}} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^4)(1 + \sqrt{x^2 + y^4 + 1})}{-x^2 - y^4} \\ &= - \lim_{(x,y) \rightarrow (0,0)} \left(1 + \sqrt{x^2 + y^4 + 1}\right) = -2 = f(0, 0).\end{aligned}$$

Q2. (a) $\nabla f = \frac{\cos(x^2 + y) - 2x^2 \sin(x^2 + y)}{yz} \mathbf{i} - \frac{xyz \sin(x^2 + y) + xz \cos(x^2 + y)}{y^2 z^2} \mathbf{j} - \frac{x \cos(x^2 + y)}{yz^2} \mathbf{k}.$

(b) Recall that the directional derivative of a function f at a point P has its maximum value in the direction of $\nabla f(P)$. At the given point we have

$$\nabla f(1, -1, 2) = \left\langle -\frac{1}{2}, -\frac{1}{2}, \frac{1}{4} \right\rangle \Rightarrow |\nabla f(1, -1, 2)| = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{16}} = \frac{3}{4}.$$

Now the desired vector is $\mathbf{u} = \frac{4}{3} \left\langle -\frac{1}{2}, -\frac{1}{2}, \frac{1}{4} \right\rangle = \left\langle -\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right\rangle.$

Q3. Let $g(x, y, z) = x^2 + y^2 + z^2$. By the method of Lagrange multipliers, the desired extreme values occur at points where ∇f and ∇g are parallel:

$$\begin{cases} \nabla f = \lambda \nabla g \\ g = 1 \end{cases} \Rightarrow \begin{cases} 2xy^2z^2 = 2\lambda x \\ 2x^2yz^2 = 2\lambda y \\ 2x^2y^2z = 2\lambda z \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

Multiplying the first equation by x , the second equation by y , and the third equation by z we obtain $\lambda x^2 = \lambda y^2 = \lambda z^2$. Note that if $\lambda = 0$, the first equation above gives $xy^2z^2 = 0$, which implies $f = 0$. On the other hand, if $\lambda \neq 0$, we obtain $x^2 = y^2 = z^2$ which with the last equation above yields $x^2 = y^2 = z^2 = \frac{1}{3}$, and in turn $f = \frac{1}{27}$. We conclude that on the sphere $x^2 + y^2 + z^2$, the minimum value of f is 0 and its maximum value is $\frac{1}{27}$.

Q4. Let C denote the described curve. Then C can be parametrized as $x = t^2/4$, $y = t$, $0 \leq t \leq 2$. Using this parametric representation, we may compute the mass m of the wire:

$$\begin{aligned}m &= \int_C \rho(x, y) ds = \int_C xy ds = \int_0^2 \frac{1}{4}t^2 \cdot t \cdot \sqrt{(t/2)^2 + (1)^2} dt = \frac{1}{16} \int_0^2 t^2 \cdot 2t \cdot \sqrt{t^2 + 4} dt \\ &= \frac{1}{16} \int_4^8 (u - 4)u^{\frac{1}{2}} du \quad (u = t^2 + 4) \\ &= \frac{1}{16} \int_4^8 \left(u^{\frac{3}{2}} - 4u^{\frac{1}{2}}\right) du = \frac{1}{16} \left[\frac{2}{5}u^{\frac{5}{2}} - \frac{8}{3}u^{\frac{3}{2}} \right]_4^8 = \frac{8(1 + \sqrt{2})}{15}.\end{aligned}$$

Q5. Let C denote the positively oriented cardioid $r = 1 - \cos \theta$, $0 \leq \theta \leq 2\pi$. By Green's Theorem, the area of the region enclosed by C is $A = \frac{1}{2} \int_C y dx - x dy$. Using the parametric equations

$$x = r \cos \theta = (1 - \cos \theta) \cos \theta, \quad y = r \sin \theta = (1 - \cos \theta) \sin \theta, \quad 0 \leq \theta \leq 2\pi$$

for C , we obtain

$$\begin{aligned}A &= \frac{1}{2} \int_0^{2\pi} \left[r \sin \theta (r' \cos \theta - r \sin \theta) - r \cos \theta (r' \sin \theta + r \cos \theta) \right] d\theta \\ &= \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{1}{2} \int_0^{2\pi} (1 - \cos \theta)^2 d\theta = \frac{3\pi}{2}.\end{aligned}$$

Q6. Suppose that the given vector field $\mathbf{F}$ has a potential function f . Then $\mathbf{F} = \nabla f$ and in particular, $f_z = 1/(x^2 + 2y^2 + 1)$ which implies $f(x, y, z) = z/(x^2 + 2y^2 + 1) + g(x, y)$. Differentiating both sides with respect to x and comparing with $f_x = -2xz/(x^2 + 2y^2 + 1)^2$ we obtain $g_x = 0$, hence $g(x, y) = h(y)$. Similarly it can be seen that $h'(y) = 0$, hence $h(y) = K$ is a constant. Thus $f(x, y, z) = z/(x^2 + 2y^2 + 1) + K$ is a potential function for $\mathbf{F}$.

$$\text{Now } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(2, 1, 4) - f(0, 0, 1) = -\frac{3}{7}.$$

Q7. The boundary curve C of S is the circle $x^2 + y^2 = 4$ in the plane $z = 4$, which can be represented by the parametric equations

$$x = 2 \cos t, \quad y = 2 \sin t, \quad z = 4, \quad 0 \leq t \leq 2\pi.$$

It is easy to see that with this parametrization, C is positively oriented with respect to S . Thus Stokes' Theorem yields

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C yz \, dx - xz \, dy + e^{xyz} \, dz \\ &= \int_0^{2\pi} (-16 \sin^2 t - 16 \cos^2 t + 0) \, dt = -32\pi. \end{aligned}$$

Q8. By the Divergence Theorem, the flux of $\mathbf{F}$ across S equals

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \text{div } \mathbf{F} \, dV = \iiint_E (y + 2y + y) \, dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 4y \, dz \, dy \, dx = \cdots = \frac{1}{6}.$$

Q9. Note that $\mathbf{r}_u = \langle -(2+\cos v) \sin u, (2+\cos v) \cos u, 0 \rangle$ and $\mathbf{r}_v = \langle -\sin v \cos u, -\sin v \sin u, \cos v \rangle$. Hence

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -(2+\cos v) \sin u & (2+\cos v) \cos u & 0 \\ -\sin v \cos u & -\sin v \sin u & \cos v \end{vmatrix} = (2+\cos v) \langle \cos u \cos v, \sin u \cos v, \sin v \rangle.$$

(a) It is easily seen that at the point $P(3, 0, 0)$ we have $\sin u = \sin v = 0$ and $\cos u = \cos v = 1$. Thus $u = v = 0$ and the normal vector at this point is $\mathbf{r}_u(0, 0) \times \mathbf{r}_v(0, 0) = \langle 3, 0, 0 \rangle$. An equation for the tangent plane to the surface S at the point P is $3(x-3) + 0y + 0z = 0$, or $x = 3$.

(b) The surface area of S is $A = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, dA$, where $D = [-\pi, \pi] \times [-\pi, \pi]$. On the other hand,

$$|\mathbf{r}_u \times \mathbf{r}_v| = (2 + \cos v) \sqrt{\cos^2 u \cos^2 v + \sin^2 u \cos^2 v + \sin^2 v} = 2 + \cos v.$$

Hence

$$A = \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} (2 + \cos v) \, dv \, du = 8\pi^2.$$